



Sergio Pérez Fernández

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https://www.linkedin.com/in/sergioperezfdez0/?locale=en_US | **YouTube:** <https://www.youtube.com/@SPDev00> | **GitHu**

b: <https://github.com/SergioPerezFdez0> | **Address:** Madrid, Spain (Home)

WORK EXPERIENCE

09/2023 – 11/2023 Madrid, Spain

SOFTWARE DEVELOPER QUENTA SOLUTIONS

Design and implementation of robust and scalable solutions using versatile languages and modern frameworks, including Python, JavaScript, and Dart.

Development of an encrypted satellite communication application for delivery trucks in the Dominican Republic.

Configuration and management of environments to ensure efficient and secure interaction between multiple Docker containers.

Team coordination and collaboration using GitHub, GitFlow, and Trello.

Achievements: Participation in an international project as part of a team using agile methodologies

EDUCATION AND TRAINING

11/2024 – 12/2024

UNITY MULTIPLAYER DEVELOPER (UNITY NETCODE FOR GAMEOBJECTS) Nathan Farrer + Gamedev.tv

Development of 3D multiplayer games using the Unity Netcode for GameObjects framework.

Creation of advanced mechanics for players and the environment with network synchronization, including movement, shooting, health, item collection, inventory, scoring, spawn points, respawning, cooldowns, and more.

Design and management of interfaces that utilize real-time and networked resources, such as a minimap, scoreboard, name tags, health bars for allies and enemies, and more.

Implementation of Relay in Unity Gaming Services to allow a player to host a game without the need to open ports or share IP addresses.

Development of a lobby system in UGS to display public game lists and enable private matches that can be joined via a code.

Integration of Multiplay Hosting in UGS to create dedicated servers as an alternative to player-hosted sessions.

Implementation of Matchmaker in UGS to support matchmaking with different queue types, such as solo play or team play with and without friendly fire.

Management of real-time interaction exclusivity to prevent conflicts with shared resources.

Implementation of local client-side action simulation while the server processes the action to enhance responsiveness.

Ability to perform Remote Procedural Calls (RPCs) to allow clients to execute specific actions without relying on the server.

Handling interactions between object layers.

Creation of particle effects optimized for a multiplayer environment.

Error and exception handling.

Website <https://www.gamedev.tv/>

01/2024 – CURRENT Madrid, Spain

MASTER'S IN VIDEO GAME DEVELOPMENT CEI

Advanced development of interactive applications in Unity.

Seamless integration of animations, including cinematics and character animation.

Configuration and optimization of AI systems, AI Navigation, and pathfinding.

Design of save systems and state recovery mechanisms.

Rendering pipeline configuration.

Application of artistic fundamentals and visual principles for character and environment design.

Creation of concept illustrations, sprites, and pixel art designs for video games.

Development of 2D, pixel art, and 3D animations, adapting to various styles.

Design of immersive interactive environments, detailed assets, UI elements, and character interfaces.
Utilization of dynamic perspectives to enhance visual storytelling.
Optimization of 3D models using advanced quad techniques in Maya.
3D modeling in ZBrush with a focus on performance.
Rigging setup and precise UV mapping for detailed texturing in Substance Painter.
Creation and application of PBR materials and custom textures for high-quality projects.
Expertise in advanced techniques to optimize visuals for games and simulations.
Development of landscapes and terrains.
Configuration of dynamic and static lighting for realistic game environments and renders.
Design and implementation of visual effects using particle systems.
Streamlined workflow for sprites, pixel art, models, terrains, materials, textures, and animations in game engines.
Collaborative experience working with artists, programmers, and designers.
Project management and version control expertise.

Website <https://cei.es/> | **Level in EQF** EQF level 5

09/2021 – 11/2023 Leganés, Spain

HIGHER TECHNICIAN IN MULTIPLATFORM APPLICATION DEVELOPMENT IES Luis Vives

Final Degree Project: Development of an online Android game based on the "dictionary game." Built with Kotlin and Jetpack Compose in Android Studio, featuring Google login, Firebase database, and an accessible design using Material Design 3. Key features include private match creation, rule customization, integrated tutorial, RAE dictionary integration, score management, and APK deployment for the Google Play Store.

Bicycle Shop Management Application: Kotlin-based microservices solution with a RESTful API, utilizing Docker, KTor, Spring Data, Spring Boot, MongoDB, and MariaDB. Features include product, order, and user role management, real-time stock and order status updates, encryption, token-based authentication, SSL security, asynchronous functions, and data validation.

Self-Service Machine Simulator: Java application for processing orders and customized tickets, with a database for managing discounts, products, and encrypted customer data. Includes JSON backup implementation and special access features for managers.

Technical Skills

Programming & Frameworks: Kotlin, Java, Python, Spring Boot, Spring Data, Mongo, Flutter, KTor.

Databases: SQL, MariaDB, MongoDB, H2, NoSQL, Odoo, Firebase.

Infrastructure & DevOps: Docker, network management, virtual machines (Windows, Linux, Ubuntu Server).

Design & Development: Scalable applications for multiple devices and operating systems, enabling real-time connectivity. Report generation and data analysis.

Security: Implementation of SSL, token-based authentication, encryption, and sensitive data protection.

Agile Methodologies: Scrum, project management, and version control.

Additional Tools: Web Services integration (SOAP, REST), web scraping, version control, flowchart design, office automation with report generation and data analysis.

Key Achievements

Development of efficient and concurrent applications using multithreading and multiprocessing.

Design of interactive and multimedia solutions for web and Android applications.

Custom ERP/CRM system development ensuring usability and data integrity.

Experience in unit testing, test planning, and deployment in production environments.

Participation in Agile-driven projects using Scrum methodology.

Website <https://www.iesluisvives.es/> |

Field of study Software and applications development and analysis , Computer use , Database and network design and administration

Level in EQF EQF level 5

2020 – 2021 Madrid, Spain

TECHNOLOGICAL BACCALAUREATE Colegio Ábaco

Website <https://colegio-abaco.com/> | **Level in EQF** EQF level 4

● **CERTIFICATIONS**

Unity Certified User: Programmer

Link https://www.credly.com/badges/8f5b004e-2a23-4ce8-beb4-9659714fa0ea/linked_in_profile

C# (Basic) Certificate

Link <https://www.hackerrank.com/certificates/26748d9d9b64>

AFFILIATIONS

Garantía Juvenil

Link <https://www.comunidad.madrid/servicios/juventud/garantia-juvenil>

LANGUAGE SKILLS

Mother tongue(s): **SPANISH**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C2	C2	C1	C2	C1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

DRIVING LICENCE

Driving Licence: B

DIGITAL SKILLS

Digital Skills - Test Results

 Information and data literacy	ADVANCED	Level 6 / 6
 Communication and collaboration	ADVANCED	Level 6 / 6
 Digital content creation	ADVANCED	Level 5 / 6
 Safety	ADVANCED	Level 6 / 6
 Problem solving	ADVANCED	Level 6 / 6

Results from [self-assessment](#) based on [The Digital Competence Framework 2.1](#)